

4. [4 marks]

(2002:09)

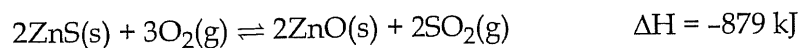
Write the equilibrium constant expression for each of the following:

[4]

Equation	$\text{CaCrO}_4(\text{s}) \rightleftharpoons \text{Ca}^{2+}(\text{aq}) + \text{CrO}_4^{2-}(\text{aq})$
Equilibrium constant expression	

Equation	$4\text{NO}_2(\text{g}) + 6\text{H}_2\text{O}(\text{g}) \rightleftharpoons 7\text{O}_2(\text{g}) + 4\text{NH}_3(\text{g})$
Equilibrium constant expression	

The following equilibrium is being investigated:

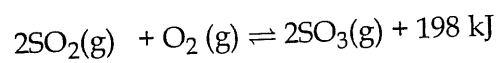


Three identical sealed boxes are set up, each containing the equilibrium mixture. Each of the boxes is treated as described below, and time is allowed for a new equilibrium to be established. In each case describe the change between the original equilibrium and the new equilibrium. [6]

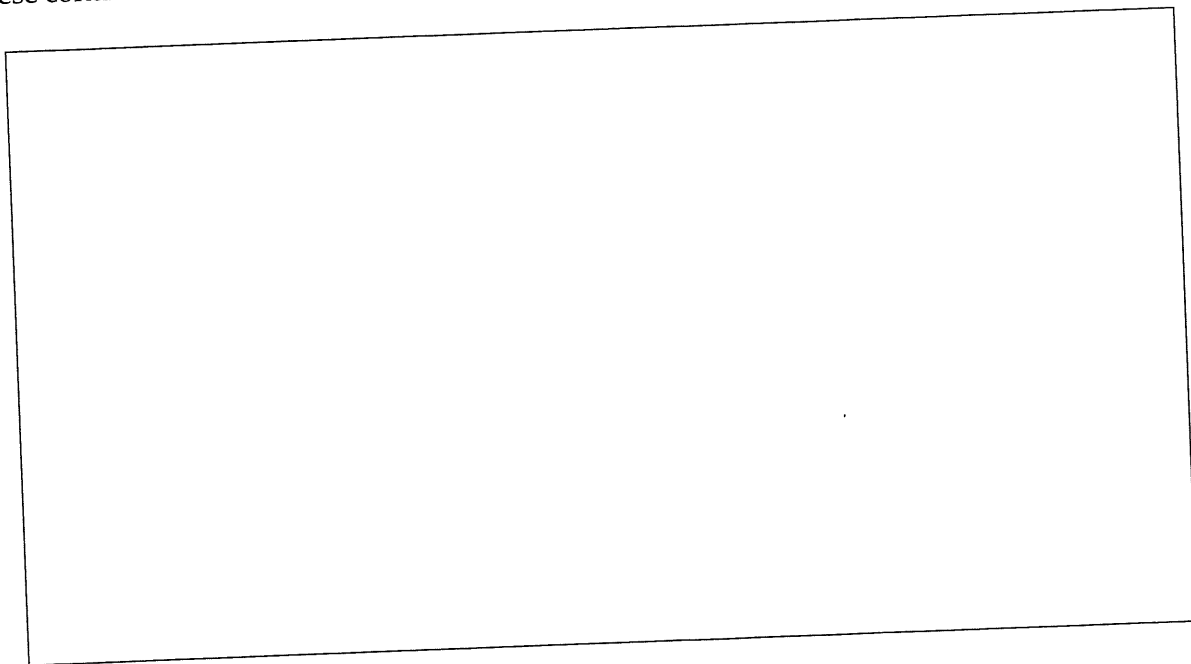
Treatment	What happens to the rate of the forward reaction? Write 'increases', 'decreases' or 'no change'.	What happens to the equilibrium position? Write 'move to the right', 'move to the left' or 'no change'.
A small amount of $\text{O}_2\text{(g)}$ is added.		
Ne(g) is pumped in, increasing the pressure of the system (no volume change).		
The reaction vessel is heated.		

6. [5 marks]

The second step in the contact process for H_2SO_4 manufacture involves the reaction



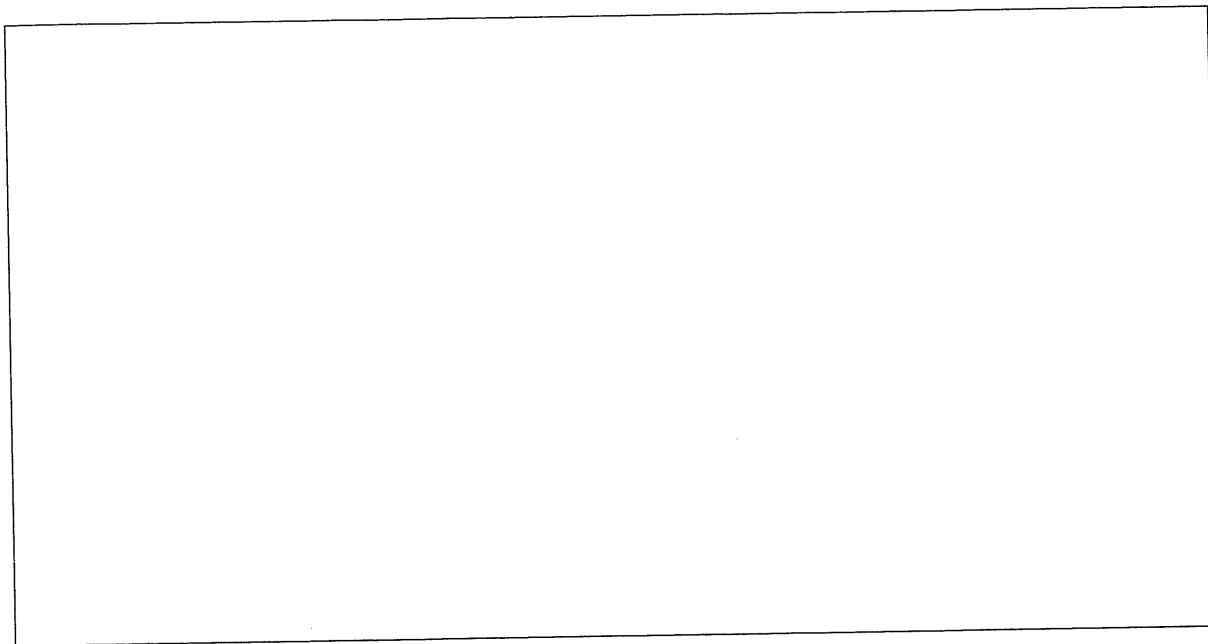
This process is carried out at 400 to 600°C and 1 atmosphere pressure. Explain the choice of these conditions.



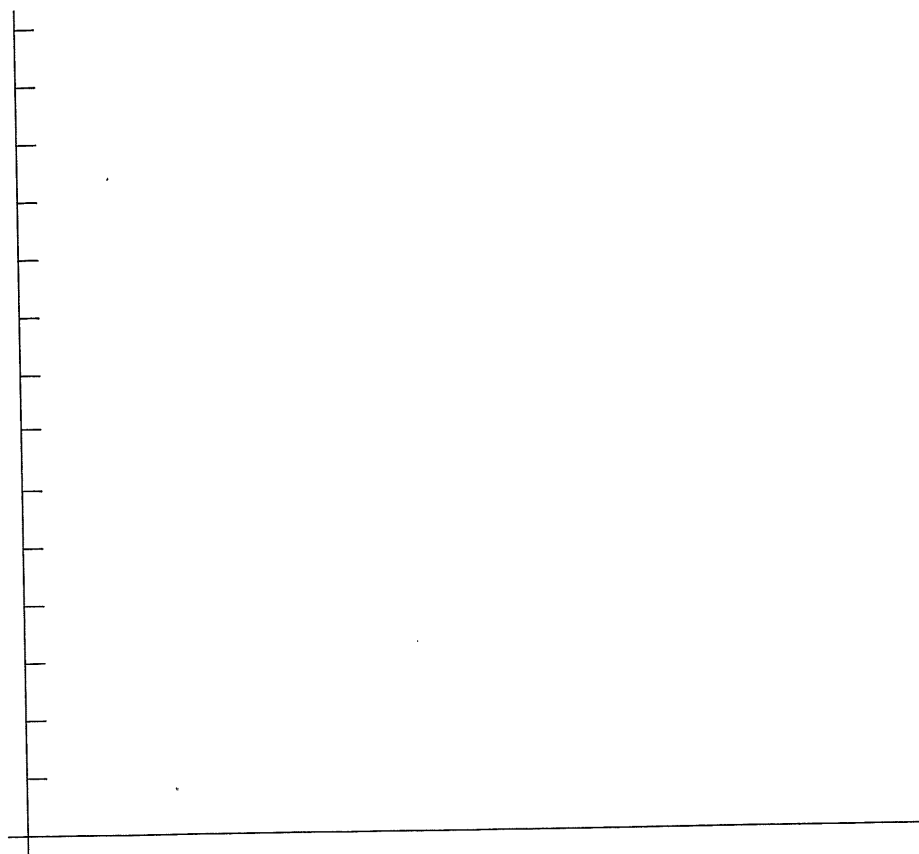
15. [8 marks]

(2004:05)

- (a) A small increase in the temperature of a reaction will often cause a significant increase in the rate of a reaction. Explain, with reference to collision theory, why this is so. Use diagrams if appropriate. [4]



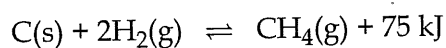
- (b) Draw a potential energy diagram for a reaction with activation energy = 50 kJ mol^{-1} and $\Delta H = + 20 \text{ kJ mol}^{-1}$. Label the diagram **and** axes well, showing the transition state, product and reactants, along with the activation energy and ΔH . [4]



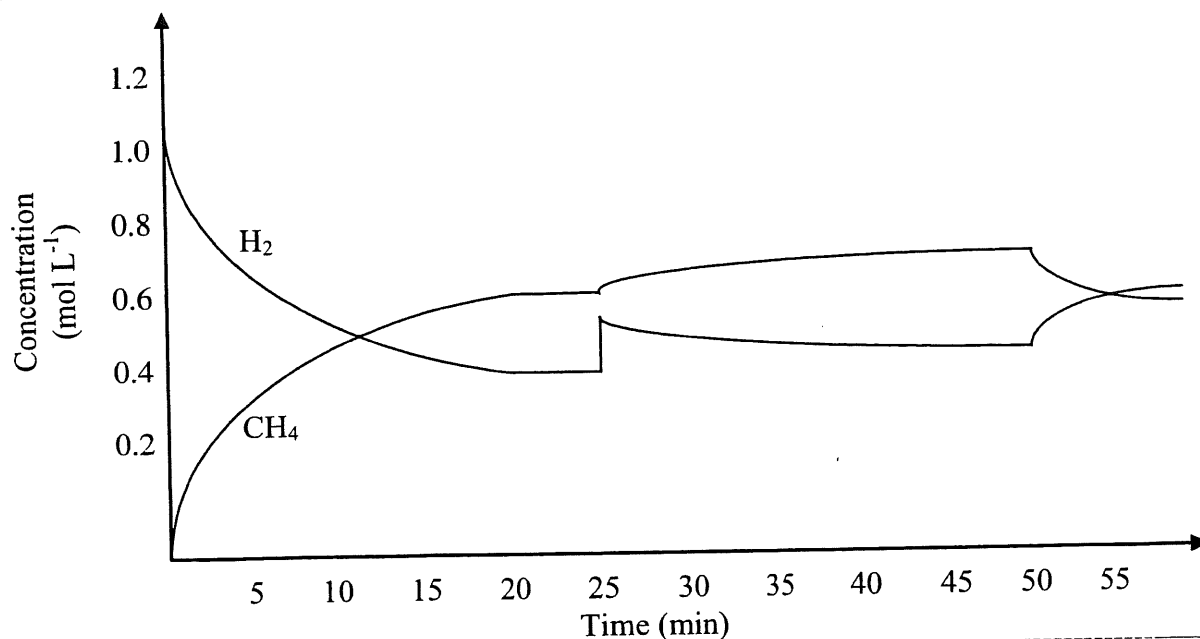
(2005:07)

21. [6 marks]

The reaction between carbon and hydrogen gas to form methane can be represented by the following equation.



The concentrations of hydrogen and methane were plotted over time and the following graph produced.



- (a) About what time was equilibrium first established? [1]
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- (b) Suggest what could have caused the change at the 25 minute mark. [1]
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- (c) Suggest what change to the system occurred at the 50 minute mark. [1]
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- (d) What would be the effect on the equilibrium if more C(s) was added to the system? [1]
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- (e) Explain, using Le Chatelier's Principle, what would be the effect of halving the volume of the reaction container. [2]
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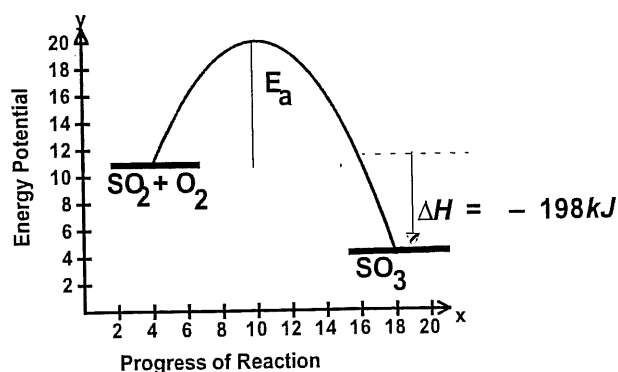
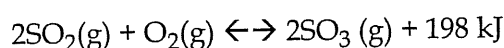
4.(2002:09) Equilibrium constant expressions for the following:

Equation	$\text{CaCrO}_4(\text{s}) \leftrightarrow \text{Ca}^{2+}(\text{aq}) + \text{CrO}_4^{2-}(\text{aq})$
Equilibrium constant expression	$K = [\text{Ca}^{2+}] [\text{CrO}_4^{2-}]$
Equation	$4\text{NO}_2(\text{g}) + 6\text{H}_2\text{O} \leftrightarrow 7\text{O}_2(\text{g}) + 4\text{NH}_3(\text{g})$
Equilibrium constant expression	$K = \frac{[\text{O}_2]^7 [\text{NH}_3]^4}{[\text{NO}_2]^4 [\text{H}_2\text{O}]^6}$

5.(2002:10) Description of the changes between the original and the new equilibrium.

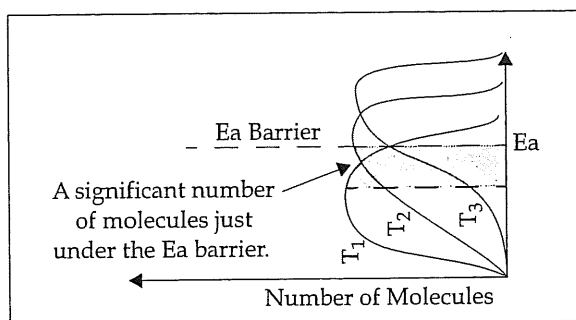
Treatment	What happens to the rate of the forward reaction?	What happens to the equilibrium position?
A small amount of $\text{O}_2(\text{g})$ is added	Increases	Moves to the right
$\text{Ne}(\text{g})$ is pumped in, increasing the pressure of the system (no volume change).	No change	No change
The reaction vessel is heated.	Increases	Moves to the left

6.(2002:11) The reasons for selecting a high temperature and a standard atmospheric pressure as conditions for the following exothermic reaction



Even though the reaction is exothermic, heating it is obviously necessary to supply the required activation energy, which is relatively high for this reaction. This high temperature undoubtedly affects the reaction yield while it has the advantage of increasing the rate. Though a slight increase in pressure would favour the products, it is considered unnecessary for economic reasons, and a pressure of 1.0 atmosphere is used.

- 15.(2004:05) a) An increase in temperature can increase the collision energy, the rate of collisions and hence the rate of effective collisions. Many of the particles which are slightly below the activation energy barrier begin to react and this increases the rate both in the forward and in the reverse direction. Note the shaded area in the graph. A significant number of molecules are just under the activation energy barrier.
- b) $T_3 > T_2 > T_1$



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Solutions

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- a) At about the 20-minute point.
- b) More hydrogen gas is pumped into the system.
- c) The temperature of the system could have been increased which has shifted the equilibrium towards the reactant side as this reaction is exothermic.
- d) None. It is a solid.
- e) Halving the volume will double the pressure. Equilibrium will shift to the right. According to the Le Chatelier's principle, an increase in pressure should move the reaction in the direction of less number of gaseous moles.